LESSON PLAN

ENGINE HEALTH MONITORING (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	02
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Identification of AN-type, NAS etc.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Identification of AN-type, NAS etc. grip length, torques, a/c bolt identification, screws, taper pins.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe the use of various standards, identification of bolts, nuts, safety methods and maintenance of cables.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of AN-type, NAS, grip length, torques, a/c bolt identification, screws, taper pins using physical samples and torque wrenches.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary difference between AN and NAS fasteners?
(b) Why is grip length critical in aircraft structural assembly?
(c) What safety risks arise from incorrect bolt identification?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• AN-type bolt head identification • NAS bolt head identification • Grip length measurement standards • Torque specifications and application • Aircraft bolt identification codes	L & Ds	PPT, IB	15 min
(b)	• Screw types and applications • Taper pin identification and usage • Safety methods for fastener installation • Torque wrench calibration checks • Marking and traceability procedures	L & Ds	PPT, PDF	20 min
(c)	• Common errors in bolt selection • Maintenance of cable fasteners • Documentation of torque values • Inspection criteria for worn pins • Storage and handling protocols	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the significance of the AN-470 series in aircraft structures.
(b) Describe the procedure for verifying taper pin integrity before installation.
(c) How do you ensure torque values are recorded correctly for maintenance logs?

12. Home Assignment and Follow Up	Prepare a brief report comparing AN and NAS fastener standards, highlighting key identification features and torque requirements for the next lesson on nuts and turn buckles.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Nuts, miscellaneous nuts, washers, Safety methods for turn buckles and various types of turn buckles.	Refer to the master syllabus reference list	LP 03

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min